Multi-Dialect Marathi Text Normalization via LLM-Driven Synthetic Augmentation

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Abstract

Low-resource regional sub-dialects face severe data scarcity and lack parallel text normalization benchmarks. In this paper, we address dialect normalization for Marathi—converting non-standard text from three major regional dialects (Southern Konkan, Northern Konkan, and Varhadi) into Standard Written Marathi. To overcome data scarcity without expensive human annotation, we introduce an automated synthetic parallel expansion pipeline combining Gemma-4 translation with a multi-tier verification engine (rule-based script filtering and LLaMA-3.1-8B semantic auditing), expanding the clean parallel corpus from 16,163 to 32,335 verified pairs. We fine-tune sequence-to-sequence transformer architectures (mT5-small and IndicBART) using task-specific normalization conditioning tags under a 5-fold cross-validation scheme. Empirical results show that mT5-small achieves state-of-the-art performance with 69.51 BLEU and 84.58 chrF++ on the expanded dataset (+12.39 BLEU over IndicBART). Crucially, multi-tier verification prevents synthetic quality collapse (+14.03 BLEU recovery over unverified data), while dataset expansion eliminates cross-dialect parameter competition.

Keywords (at least 4-5): Dialect Normalization, Low-Resource NLP, Marathi Dialects, Sequence-to-Sequence Modeling, Synthetic Data Augmentation, Task-Specific Conditioning, mT5, IndicBART.

1 Introduction

Rapid development of deep learning architectures and self-attention mechanisms [1] has made it possible to develop systems capable of understanding and generating natural language. Open-weight models have gained popularity to process text in various domains as they provide a finer control over system architecture, runtime environment, and training data [2]. Yet current language systems tend to work efficiently with standard high-resource languages. Queries based on non-standard input yield unexpected and undesired outputs. [3]. Speakers of regional dialects face many issues in systems that use these models across automated speech recognition, machine translation and other language processing systems [4, 5].

Dialect normalization is the task of translating non-standard regional dialects into standard written language. The main challenge te be adddressed while designing such systems is the preservation of the meaning of original text. [6]. It serves to be an essential layer for digital linguistic inclusion. Countries having large and diverse speech communities, such as India, which has 22 official languages and hundreds of regional dialects spoken by 1.4 billion people, regional varieties have remaind unattended in computational linguistic research. An example of such linguistic disparity is Marathi, an Indo-Aryan language spoken by over 83 million people in western India [7].

Development reliable pipelines for regional Indic dialects has faced these fundamental challenges:

1. **Resource Scarcity**: Parallel corpora mapping the regional dialects to standard written language do not exist. Curation of such sets by experts fluent in multiple dialects is both

expensive and unscalable [6, 8].

2. **Morphological and Dialectal Heterogeneity**: Marathi exhibits rich morphology where different words undergo varied complex transformations across different regions where the dialects are spoken [7, 9]. Consequently, evaluations vary significantly across individual dialects—ranging from systematic suffix alternation in Varhadi to intense lexical and phonetic divergence in Northern and Southern Konkan [7, 10].
3. **Underperformance of Multilingual Models**: Performance of pre-trained models like IndicBART [11], IndicNLP Suite [12], and mT5 [13] seems to drop drastically on non-standard language variants due to subword fragmentation during tokenisation and out-of-vocabulary phonological shifts [3, 14].

Our work mainly tries to investigate and conclude 1. how do normalisation scores vary across regional Marathi dialects which have differnet degrees of variations as compared to standard marathi; 2. how effectively can existing sequence-to-sequence architectures adapt to dialect-to-standard mapping under task-specific conditioning; and 3. whether automated synthetic data expansion can accurately solve the problem of data scarcity and eliminate cross-dialect hallucinations and grammatical errors [15, 16]. To address these questions, we construct an end-to-end normalization framework combining LLM-based synthetic generation, multi-tier syntactic and semantic verification, and conditioned transformer fine-tuning across both single-dialect and multi-dialect settings.

Our core contributions are:

- **Parallel Corpus Generation**: Based on transcripts from the RESPIN dataset [17], we construct a parallel corpus for three of the Marathi dialects (Southern Konkan, Northern Konkan, and Varhadi) mapped to Standard Marathi .
- **Automated Synthetic Augmentation with Verification**: We design a data expansion pipeline paired with a multi-tier verification engine , which has rule-based script filtration and LLM-based semantic inspection.

2 Related Work

2.1 Regional Dialect Standardization across Language Families

Early approaches in the context of European languages used Character-Level Statistical Machine Translation (CSMT) and phrase-based models for Swiss German Dialects [18] which were later replaced by pre-trained language models [19, 20]. Similar approaches have been developed for Finnish, Swedish, and Estonian dialects [21–23]. A study of four language families by Kuparinen et al. [24] suggested that fine-tuned sequence-to-sequence transformers surpassed the rule-based and SMT baselines. Other than European countries, East Asian and African-Asian varieties have also shown similar patterns. Japanese regional dialects have been processed using phrase-level LSTM modules [25], while Mandarin scripts relied on acoustic-phonetic mapping [26]. The Conventional Orthography for Dialectal Arabic (CODA) framework [27] in the landscape of Arabic languages gave guidelines for Modern Standard Arabic (MSA) which led to multi-city translation benchmarks (MADAR [28], NADI [29]) and morphology-oriented prompt design for LLMs [30, 31].

2.2 Indic Dialect Landscape and Marathi NLP

Even though pre-trained models such as IndicBART [11] and mT5 [13] which are trained on multilingual corpora cover major languages, they lack understanding of regional sub-dialects. Models based on a single language like L3Cube-MahaNLP [32] offer pre-trained marathi models for classification, but do not support generative text standardization. INDIC-DIALECT [14] considers Hindi varieties (Bhojpuri, Magahi, Maithili), for evaluation via WFST frameworks for text normalization while preserving the underlying semantics [33]. However, Marathi dialects were primarily dealt for acoustic/ASR classification [34, 35], or zero-shot LLM retrieval [36, 37].

Table 1: RESPIN-S1.0 Marathi Source Corpus Statistics

Code	Dialect Name	Utterances (%)	Duration (h)	Speakers	Unique Prompts
D1	Southern Konkan (Malvani)	203,651 (25.14%)	247.31	732	5,838
D2	Northern Konkan (Ahirani)	198,560 (24.52%)	265.25	557	5,825
D3	Standard Marathi (Puneri)	208,850 (25.79%)	255.79	608	6,077
D4	Varhadi (Vidarbha)	198,873 (24.55%)	257.70	747	5,330
Total	**All Dialects**	**809,934 (100%)**	**1,026.06**	**2,644**	**23,069**

Our work fills this gap by presenting an end-to-end neural fine-tuning pipeline with evaluations and data augmentation strategies.

2.3 LLM Distillation and Multi-Tier Verification

LLMs can be used to suffice for data scarcity in various tasks [38–40]. Knowledge distillation strategies can be employed to train compact and efficient models based on the outputs generated by a larger teacher model [15, 41, 42]. But the use of such techniques for low-resource regional dialects involves significant amount of hallucinations and noise [15, 16]. We solve this by utilising Gemma-4 with a robust morphologically aware system prompt and a multi-tier verification system that ensures integrity of the data generated.

3 Methodology

3.1 Target Dialects and Linguistic Properties

We target the following major Marathi dialects that span the primary regions of Maharashtra and exhibit distinct linguistic and phonetic patterns based on the RESPIN corpus:

1. **Southern Konkan (Konkan Coast)**: Characterized by shortening of final-word vowels, softening of dental consonants, and different auxiliary verb forms. Lexical differences include systematic vowel shifts: standard जातो ("he goes") becomes Southern Konkan जातां; standard कुठे ("where") becomes कुठंय. Domain-specific terminology generally remains untouched but often accompanied by Konkani words.
2. **Northern Konkan (Khandesh Region)**: Heavily influenced by neighboring Gujarati and Bhili language families, thus it exhibits significant consonant simplification and dental consonant shifts. Notable transformations include: standard कशाला ("why") $\rightarrow$ Northern Konkan कसाले; standard मुलगा ("boy") $\rightarrow$ Northern Konkan पोर. Northern Konkan possesses the highest degree of lexical divergence among the three target dialects.
3. **Varhadi (Vidarbha/Eastern Maharashtra)**: Spoken in Vidarbha, it exhibits characteristic differences and verb suffix modifications. Standard interrogative का? ("why?") becomes Varhadi काऊन?; verbal forms undergo regular suffix mapping patterns. Among the three dialects, Varhadi maintains closest structural similarity to Standard Marathi as transformations are more systematic.

3.2 Original Corpus Curation and Foundation in RESPIN

Base sentences for our task are drawn from the domain-specific transcripts of the RESPIN-S1.0 initiative [17], developed at IISc Bangalore under the Gates Foundation project. While RESPIN has audio recordings and their transcripts across agriculture and banking domains for Indian languages and their dialects, it lacks aligned parallel dialect-to-target language translations required for efficient normalisation.

The large speaker diversity ensures broad coverage of intra-dialect phonetic and lexical variation, while the balanced gender and domain splits reduce systematic bias in our derived parallel corpus.

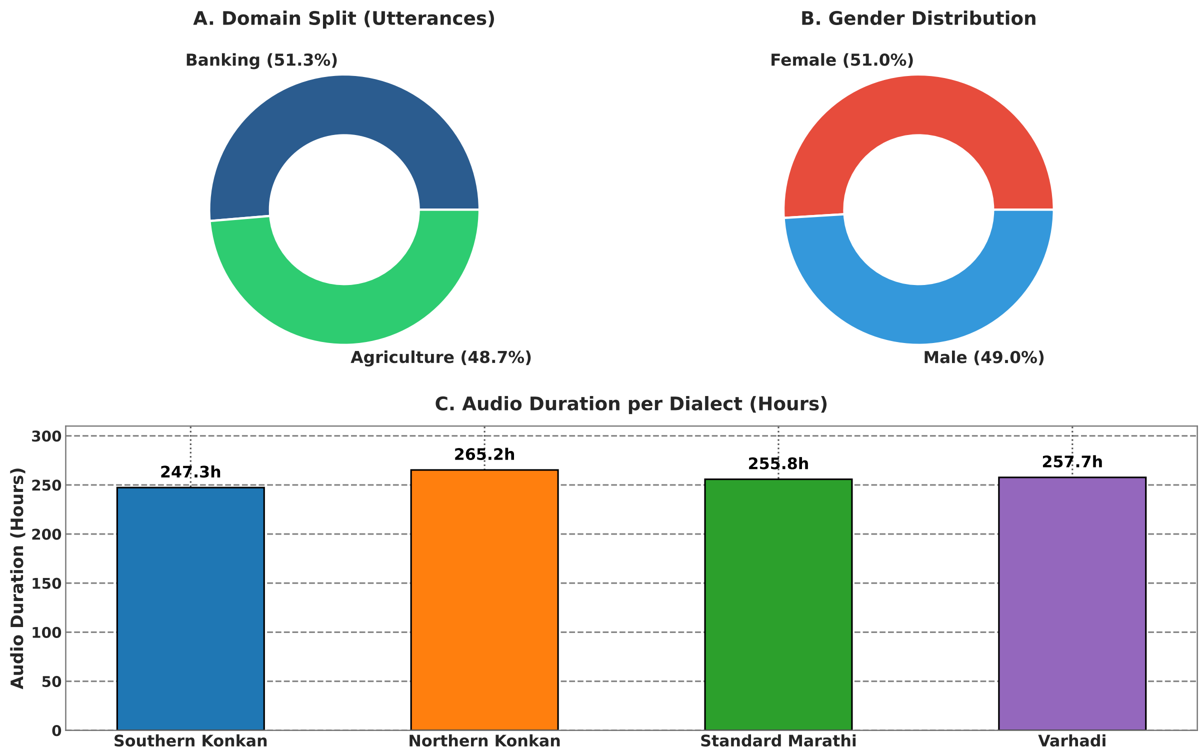

Figure 1: RESPIN-S1.0 Marathi source corpus demographics: (A) domain distribution; (B) gender split; (C) audio recording duration per dialect variant.

3.3 LLM-Driven Translation and Verification Pipeline

To deal with data scarcity, we developed an automated synthetic data translation and generation pipeline powered by Gemma-4 [43] in 9B-instruction-tuned (gemma-4) configurations, accessed via Ollama. We considered several design decisions to ensure generation quality to be maximised:

- **1-Prompt-Per-Sentence Architecture**: Instead of generating batches of parallel pairs per API call (which risks array length mismatches and context contamination), each dialectal sentence is processed separately with a self-contained prompt.
- **Detailed Prompting Strategy**: The generation prompt enforces semantic fidelity rules: the standard output must preserve exact meaning, domain terminology (agriculture/banking), named entities, and numerical values from the input. The prompt includes dialect-specific lexical mapping tables and 5-8 few-shot examples per dialect.
- **Temperature Control**: Generation temperature is set to 0.3 with top-p = 0.9 to balance diversity with fidelity.

All generated pairs undergo a three-tier verification and correction pipeline:

1. **Tier 1: Rule-Based Syntactic Filter.** Automated validation checks implement Indic Abugida script normalization principles [9], including:
   - *Devanagari Script Integrity*: Both source and target must possess $> 90\%$ Devanagari Unicode characters within the `U+0900--U+097F` range, automatically discarding unnecessary characters.
   - *Length Ratio Constraint*: $0.5 \leq |L_{\text{dialect}}|/|L_{\text{standard}}| \leq 2.0$ to avoid hallucinations.
   - *Non-Identity Constraint*: $L_{\text{dialect}} \neq L_{\text{standard}}$ to exclude identical string copies.
   - *Minimum Token Count*: Both source and target must contain ≥ 3 whitespace-separated tokens.
2. **Tier 2: LLM Semantic Verification.** A secondary LLM evaluator using `llama-3.1-8b-instant` (accessed via Groq API) performs semantic equivalence verification. The verifier rates

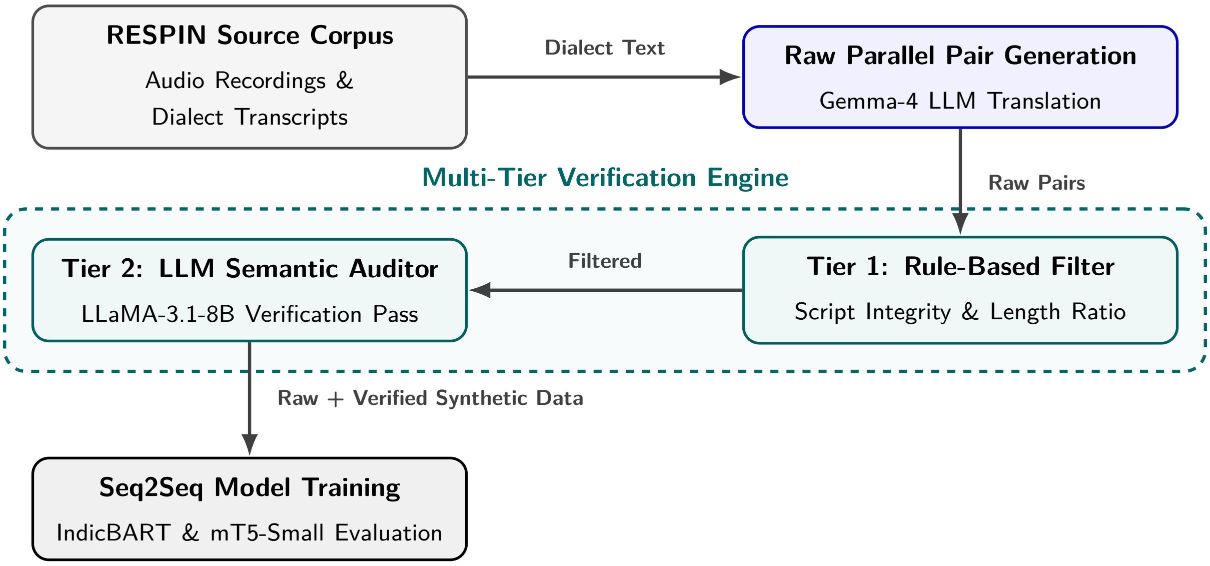

Figure 2: System architecture of the multi-dialect normalization framework.

Table 2: Statistics based on generated and verified data

Dialect Name	Original Clean Pairs	Raw Synthetic Generated	Clean Synthetic Verified	Corrupted / Rejected Pairs (%)	Total Clean Expanded Pairs
Southern Konkan	5,576	5,838	5,569	269 (4.61%)	**11,145**
Northern Konkan	5,501	5,825	5,534	291 (5.00%)	**11,035**
Varhadi	5,086	5,330	5,069	261 (4.90%)	**10,155**
Total	**16,163**	**16,993**	**16,172**	**821 (4.83%)**	**32,335**

on a 1–5 score scale considering: (a) semantic preservation; (b) absence of hallucinated entities; (c) absence of Hindi/English contamination; and (d) complete standardization of dialects. Pairs scoring < 4.0 are rejected or routed to correction.

3. **Closed-Loop Correction Engine.** Pairs flagged by Tier 1 or Tier 2 enter a two-step correction loop: Step 1 generates corrected translations using specialised prompts based on the pre-defined reason of identified failure.

This leaves us with a total of 16,163 pairs

3.4 Synthetic Data Expansion

After construction the original 16,163 verified parallel pairs through the RESPIN corpus, we applied the same pipeline to synthetically expand it. The expansion phase uses a similar system prompt, inspired from that of translation phase to generate additional pairs from scratch that almost doubles the data size.

The pipeline was used to generate parallel pairs from the RESPIN transcripts and for synthetic data expansion. Out of the 16,993 synthetic parallel pairs generated by the LLM pipeline, our multi-tier verification engine flagged and discarded 821 flawed pairs (an overall corruption/rejection rate of 4.83%), yielding 16,172 clean verified pairs as mentioned in Table 2. Combining with the 16,163 clean original pairs, the final expanded dataset consists of 32,335 verified parallel sentence pairs across all three dialects.

3.5 Sequence-to-Sequence Model Conditioning and Fine-Tuning Setup

To convert the pre-trained seq2seq language architectures into dedicated dialect normalizers, models are fine-tuned under a 5-fold cross-validation scheme on an 85/15 stratified train-test split. Models were conditioned using specific tags attachaed before the actual text.

- **Task Normalization Prefixes**: A normalisation tag (`<normalize>`) is attached that instructs the encoder to perform the dialect-to-standard text transformation.
- **Dialect-Source Identification Tags**: Source sentences are prefixed with dialect markers

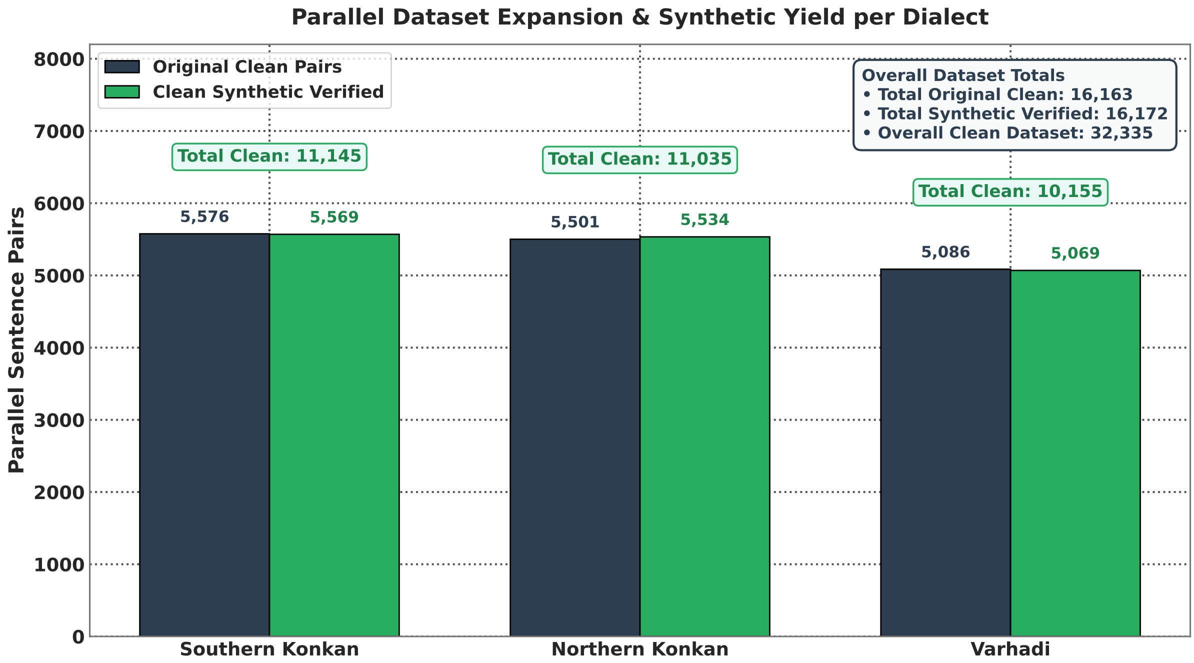

Figure 3: Parallel dataset expansion composition per dialect partition, comparing original parallel pairs against synthetic verified pairs produced by the LLM generation and auditing pipeline.

Table 3: Detailed Architecture Comparison: IndicBART (244M) vs. mT5-small (300M)

Property	**IndicBART (244M)**	**mT5-small (300M)**
Base Architecture	mBART-50 (BART Enc-Dec)	T5 Encoder-Decoder
Position Encoding	Absolute sinusoidal	Relative positional bias
Vocabulary Size	64,000 SentencePiece	250,000 SentencePiece (mC4)
Pre-training Data	11 Indic + English	101 languages (mC4)
Script Approach	Devanagari-unified	Native multi-script
Target Conditioning	Required <2mr> token	Raw text-to-text (no prefix)
Learning Rate	5×10^{-5} (linear warmup)	5×10^{-4} (AdaFactor)

,<D1> for Southern Konkan, <D2> for Northern Konkan, and <D4> for Varhadi for explicitly conditioning the model partiiculary the in multi-dialect training scenario.

- **Target Standardization Tags**: Decoder target conditioning is done using standard language tags (<standard_mr>), enforcing the output alignment into formal Standard Written Marathi.

Table 3 details the underlying architectural properties and hyperparameter configurations for IndicBART (244M) and mT5-small (300M).

3.6 Evaluation Protocol

Even though we follow a standard train/test split strategy for stable performance evaluations, the final reports were generated on the official held-out test set provided by RESPIN which had around 2,170 utterances across target dialects after training the model on the complete train set. This setup is applied across all dialects on the original transcript-based data and their synthetically expanded versions. Results are reported using metrics that are evaluated using SacreBLEU [44] and Word Error Rate (WER) scripts:

- **BLEU** (Bilingual Evaluation Understudy): BLEU-4 with 4-gram precision, brevity penalty, with an exponential smoothing method.
- **chrF++**: Character n-gram F-score combined with word unigrams and bigrams.

Table 4: Comprehensive Evaluation Matrix on `IISc_RESPIN_test_mr` Benchmark Set (2,170 Utterances)

Model Scope	Training Data	Dialect / Subset	Utts	IndicBART (244M)		mT5-Small (300M)	
				BLEU (↑)	WER (↓)	BLEU (↑)	WER (↓)
Single-Dialect	Original	Southern Konkan	559	**57.80**	**26.60%**	43.86	34.37%
		Northern Konkan	540	**90.13**	**6.46%**	79.46	11.95%
		Varhadi	516	**83.59**	**10.19%**	74.81	14.73%
	Synthetically Expanded	Southern Konkan	559	24.06	60.32%	**44.36**	**32.75%**
		Northern Konkan	540	65.41	28.70%	**79.76**	**12.61%**
		Varhadi	516	67.97	25.43%	**76.90**	**14.19%**
Multi-Dialect	Original	Southern Konkan	559	26.08	52.98%	**40.94**	**35.34%**
		Northern Konkan	540	47.92	32.64%	**77.50**	**14.65%**
		Standard Benchmark	555	96.12	3.12%	**96.65**	**1.91%**
		Varhadi	516	73.04	26.96%	**73.47**	**16.16%**
		Overall Benchmark	**2,170**	62.98	30.13%	**72.18**	**17.23%**
	Synthetically Expanded	Southern Konkan	559	**52.15**	36.50%	43.88	**34.96%**
		Northern Konkan	540	54.35	25.65%	**78.16**	**13.55%**
		Standard Benchmark	555	96.80	1.65%	**97.39**	**1.37%**
		Varhadi	516	69.96	21.04%	**74.74**	**15.57%**
		Overall Benchmark	**2,170**	**76.50**	**16.23%**	73.48	16.58%

Table 5: Benchmark Performance Matrix on Original Datasets: **Single-Dialect** Models (5-Fold CV)

Dialect Name	Total Pairs	Test Set (IB / mT5)	IndicBART (244M)		mT5-Small (300M)	
			BLEU	chrF++	BLEU	chrF++
Southern Konkan (D1)	5,576	837 / 836	**48.54**	**78.35**	46.31	74.43
Northern Konkan (D2)	5,501	826 / 825	**60.58**	**80.30**	60.31	77.34
Varhadi (D4)	5,086	763 / 762	76.63	90.93	**81.00**	**91.21**

- **Normalized WER / CER**: Word Error Rate and Character Error Rate after Devanagari Unicode script standardization.

4 Findings

4.1 Comprehensive Test Benchmark on `IISc_RESPIN_test_mr`

Key observations from Table 4 and Figure 4 include:

1. **mT5-Small Performance**: `mT5-small` upon finetuning on the synthetically expanded dataset achieves 73.48 BLEU and 87.70 chrF++ on the overall test set.
2. **IndicBART Scaling**: Fine-tuning IndicBART on the synthetically expanded multi-dialect corpus achieves **76.50 BLEU** and **88.04 chrF++**, suggesting that larger data size brings up the efficiency in mBART-50 architectures.

4.2 **Single-Dialect** vs. **Multi-Dialect** Benchmark Results on Original and Synthetic Datasets

Table 5 presents the performance matrix on original single-dialect benchmarks, and Table 6 explains the dialect-wise breakdown of the multi-dialect model on original data.

Table 7 presents the performance matrix on synthetically expanded single-dialect benchmarks, and Table 8 details the dialectwise evaluation breakdown of the multi-dialect model on expanded data.

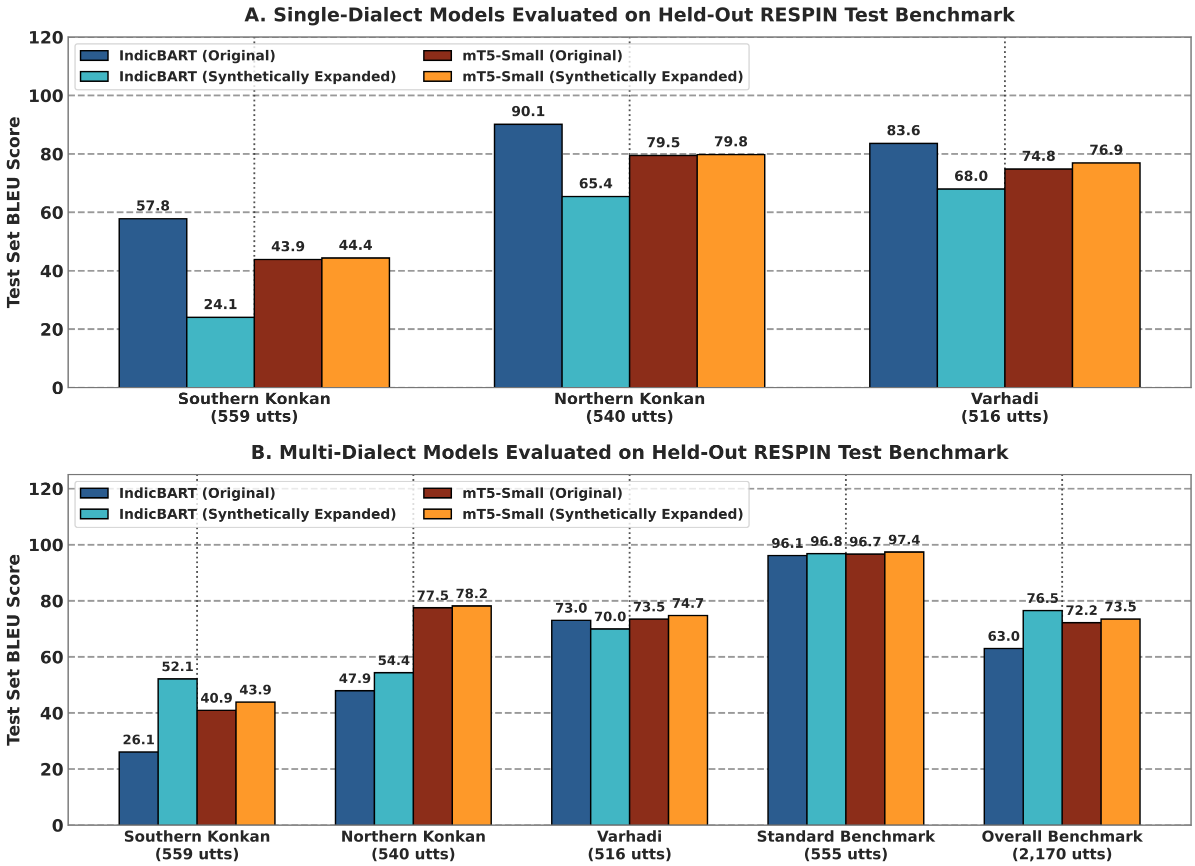

Figure 4: Comprehensive sequence-to-sequence evaluation matrix on the held-out RESPIN test benchmark (2,170 utterances), comparing IndicBART (244M) and mT5-small (300M) across single-dialect and multi-dialect training configurations.

4.3 Impact of Synthetic Data Augmentation

Table 9 isolates the effect of dataset scaling by comparing performance across original and synthetically expanded corpora for each model architecture. Overall, multi-dialect models exhibit significant net gains (+8.95 BLEU for IndicBART, +6.22 BLEU for mT5-small), confirming that data scarcity was a primary performance bottleneck.

Analyzing Table 9 through data availability and model architecture dimensions clarifies these dynamics:

- **Data Availability & Dialect Sparsity**: Southern Konkan (D1) suffered from extreme baseline data sparsity (26.21 BLEU for IndicBART, 48.28 BLEU for mT5-small). Doubling clean training pairs via synthetic expansion provided massive performance gains (+25.85 BLEU for IndicBART, +18.34 BLEU for mT5-small) by supplying critical subword affix mappings. Conversely, Varhadi (D4) already possessed a saturated baseline (72.75–79.57 BLEU), where synthetic expansion caused a minor precision drop (-2.48 and -0.84 BLEU) due to LLM-generated suffix over-generalization slightly diluting exact N-gram matches against gold references.
- **Model Architecture Differences**: mT5-small consistently outperforms IndicBART across both original (63.29 vs. 48.17 BLEU) and expanded (69.51 vs. 57.12 BLEU) combined benchmarks. This architectural superiority stems from: (1) **Subword Vocabulary Capacity**: mT5-small's 250,000-token SentencePiece vocabulary (mC4) segments non-standard dialect affixes with lower fragmentation than IndicBART's 64,000 Devanagari-unified vocabulary; and (2) **Relative Positional Embeddings**: mT5's relative positional bias models agglutinative dialectal suffix merges far more effectively than IndicBART's absolute sinusoidal encodings.

Table 6: Benchmark Performance Matrix on Original Datasets: **Multi-Dialect** Dialectwise Breakdown (5-Fold CV)

Evaluated Subset	**Total Pairs**	**Test Set (IB / mT5)**	**IndicBART (244M)**		**mT5-Small (300M)**	
			BLEU	**chrF++**	**BLEU**	**chrF++**
Southern Konkan (D1)	16,163	837 / 836	26.21	53.15	**48.28**	**75.04**
Northern Konkan (D2)	16,163	826 / 798	47.59	67.16	**62.35**	**78.92**
Varhadi (D4)	16,163	763 / 790	72.75	85.63	**79.57**	**90.51**
Overall Combined	**16,163**	**2,426 / 2,424**	**48.17**	**68.58**	**63.29**	**81.37**

Table 7: Benchmark Performance Matrix on Synthetically Expanded Datasets: **Single-Dialect** Models (5-Fold CV)

Dialect Name	**Total Pairs**	**Test Set (IB / mT5)**	**IndicBART (244M)**		**mT5-Small (300M)**	
			BLEU	**chrF++**	**BLEU**	**chrF++**
Southern Konkan (D1)	11,145	1,672 / 1,671	47.25	64.69	**65.10**	**82.88**
Northern Konkan (D2)	11,035	1,656 / 1,655	40.51	62.21	**62.07**	**79.29**
Varhadi (D4)	10,155	1,524 / 1,523	73.62	86.75	**78.89**	**90.59**

4.4 Verification Engine Impact: Raw Unverified vs. Filtered Clean Data

Performance of our verification and filtering engine is measured by conducting a comparative analysis on models trained on *Raw Unverified Synthetic Data* versus *Filtered Clean Synthetic Data* (the expanded suite). The raw unverified dataset contains 19,914 parallel pairs, that included 3,742 rejected pairs that contained garbage translations. Table 10 presents the quantitative comparison on the benchmark suite.

Analyzing Table 10 through data quality and architectural resilience dimensions yields three key insights:

1. **Data Quality & Error Propagation**: Training on raw unverified synthetic data (19,914 pairs) causes severe performance degradation (-14.03 BLEU for IndicBART, -8.30 BLEU for mT5-small). Unfiltered LLM outputs contain hallucinated stems, script noise, and spurious suffix rules, which inject corrupted gradient signals during backpropagation. Multi-tier verification (rule-based script filtering and LLM semantic auditing) removes 3,742 flawed pairs while expanding clean data to 32,335 pairs, successfully recovering BLEU scores to 57.12 (IndicBART) and 69.51 (mT5-small).
2. **Architectural Resilience to Synthetic Noise**: IndicBART exhibits extreme sensitivity to synthetic noise, suffering a 24.6% relative BLEU drop on unverified data ($57.12 \rightarrow 43.09$). In contrast, mT5-small demonstrates superior structural resilience with a smaller relative drop of 11.9% ($69.51 \rightarrow 61.21$). This resilience is directly attributed to mT5's larger 300M parameter capacity and extensive 101-language pre-training, which provides stronger regularization against noisy subword alignment gradients.
3. **Essential Role of Multi-Tier Auditing**: Combining rule-based orthographic checks with LLM-based semantic verification ensures that synthetic data augmentation scales quantitative performance without introducing systematic morphological drift.

4.5 Model Architecture Analysis

As shown in Table 3, the 69.51 vs. 57.12 BLEU gap on the expanded multi-dialect task (+12.39 absolute) can be attributed to three key architectural differences:

1. **Vocabulary Coverage**: mT5's 250K vocabulary provides 3.9$\times$ more subword units than IndicBART's 64K, enabling finer-grained tokenization of morphologically complex Marathi

Table 8: Benchmark Performance Matrix on Synthetically Expanded Datasets: **Multi-Dialect** Dialect-wise Breakdown (5-Fold CV)

Evaluated Subset	Total Pairs	Test Set (IB / mT5)	IndicBART (244M)		mT5-Small (300M)	
			BLEU	chrF++	BLEU	chrF++
Southern Konkan (D1)	32,335	1,672 / 1,710	52.06	72.15	**66.62**	**83.57**
Northern Konkan (D2)	32,335	1,656 / 1,627	49.08	69.90	**62.72**	**79.63**
Varhadi (D4)	32,335	1,524 / 1,513	70.27	84.27	**78.73**	**90.46**
Overall Combined	**32,335**	**4,852 / 4,850**	**57.12**	**75.49**	**69.51**	**84.58**

Table 9: Impact of Synthetic Data Augmentation: Original vs. Synthetically Expanded Data BLEU Change (5-Fold CV)

Evaluated Subset	IndicBART BLEU			mT5-Small BLEU		
	Original	Expanded	Δ	Original	Expanded	Δ
Southern Konkan (D1)	26.21	52.06	**+25.85**	48.28	**66.62**	**+18.34**
Northern Konkan (D2)	47.59	49.08	**+1.49**	62.35	**62.72**	**+0.37**
Varhadi (D4)	**72.75**	70.27	-2.48	**79.57**	78.73	-0.84
Overall Combined	**48.17**	**57.12**	**+8.95**	**63.29**	**69.51**	**+6.22**

dialectal forms. This is critical for dialect normalization, where subtle suffix and infix changes must be captured at the subword level.

2. **Positional Encoding**: mT5 uses relative positional biases, which generalize better to variable-length sequences typical of dialectal text where sentence length may differ substantially between dialect and standard forms.
3. **Conditioning Overhead**: IndicBART requires explicit language tokens (`<2mr>`) that add conditioning complexity, whereas mT5's pure text-to-text format enables direct mapping without task-specific architectural constraints.

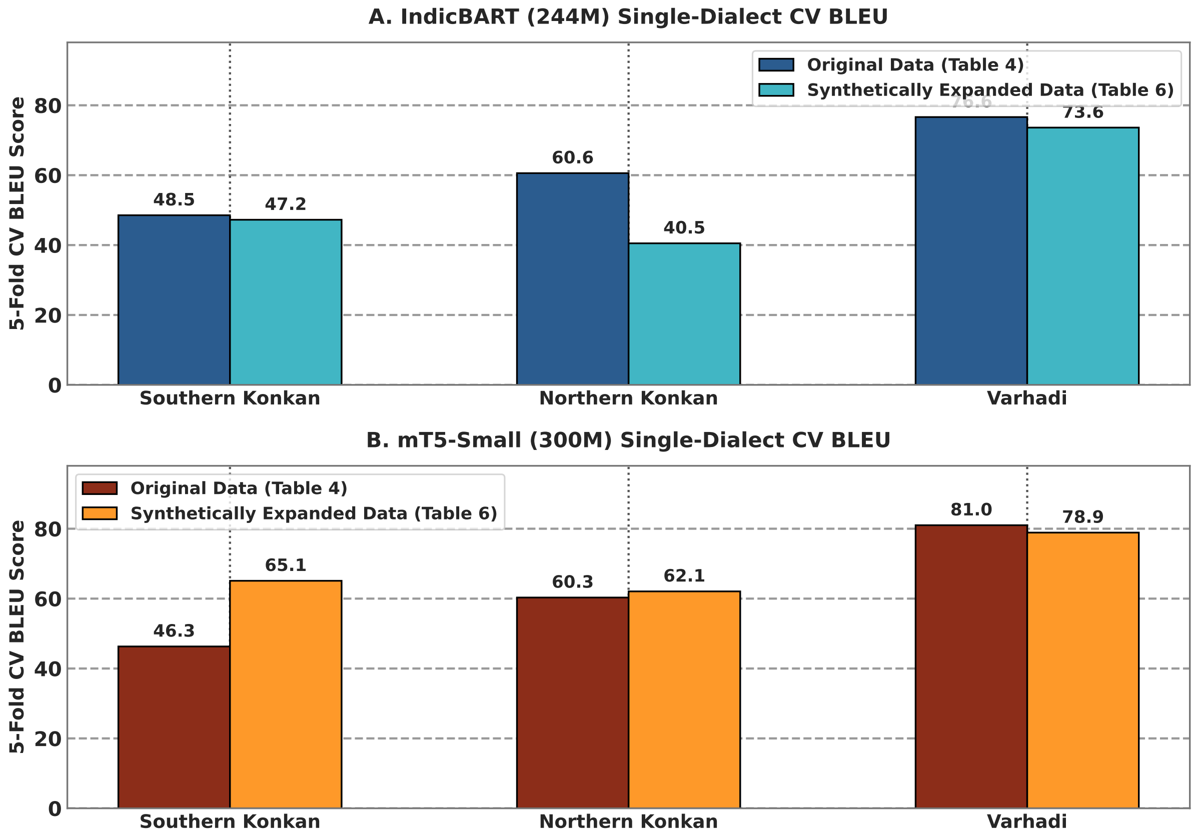

Figure 5: Single-dialect 5-fold cross-validation performance shift comparing Original and Synthetically Expanded models for IndicBART (244M) and mT5-small (300M).

Table 10: Impact of Multi-Tier Verification Engine: Raw Unverified vs. Filtered Clean Synthetic Data

Model	**Pipeline State**	**Total Pairs**	**BLEU**	**chrF++**	**$\triangle$ BLEU**	**$\triangle$ chrF++**
IndicBART (244M)	Raw Unverified Data	19,914	43.09	64.54	−14.03	−10.95
	Filtered Clean Data	32,335	**57.12**	**75.49**	**+14.03**	**+10.95**
mT5-Small (300M)	Raw Unverified Data	19,914	61.21	80.18	−8.30	−4.40
	Filtered Clean Data	32,335	**69.51**	**84.58**	**+8.30**	**+4.40**

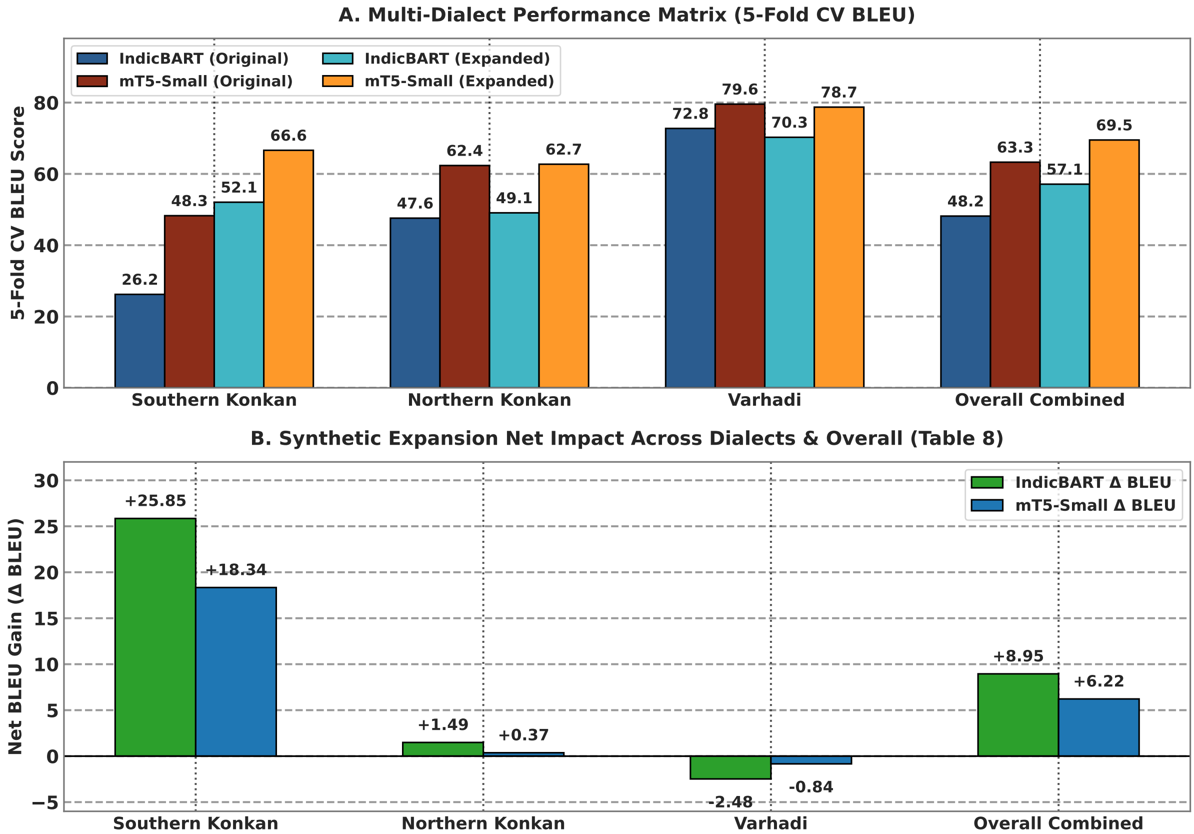

Figure 6: Multi-dialect 5-fold cross-validation benchmark comparison: (A) Absolute BLEU scores across model architectures and corpora partitions, and (B) Net synthetic expansion impact (Δ BLEU across sub-dialects).

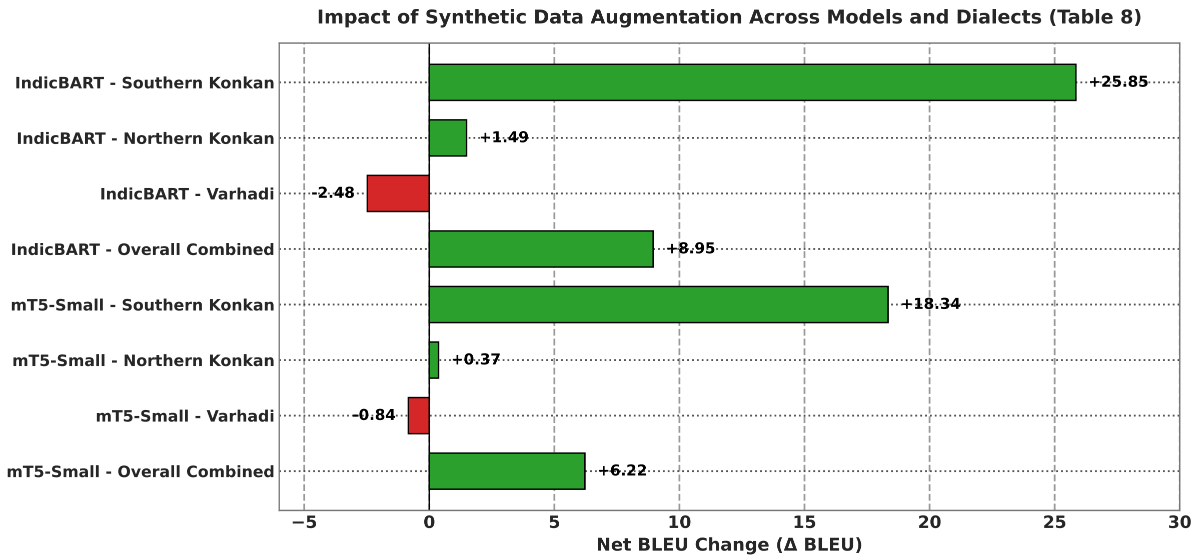

Figure 7: Impact of LLM synthetic data expansion (Δ BLEU score comparing synthetically expanded vs. original 5-fold cross-validation benchmarks).

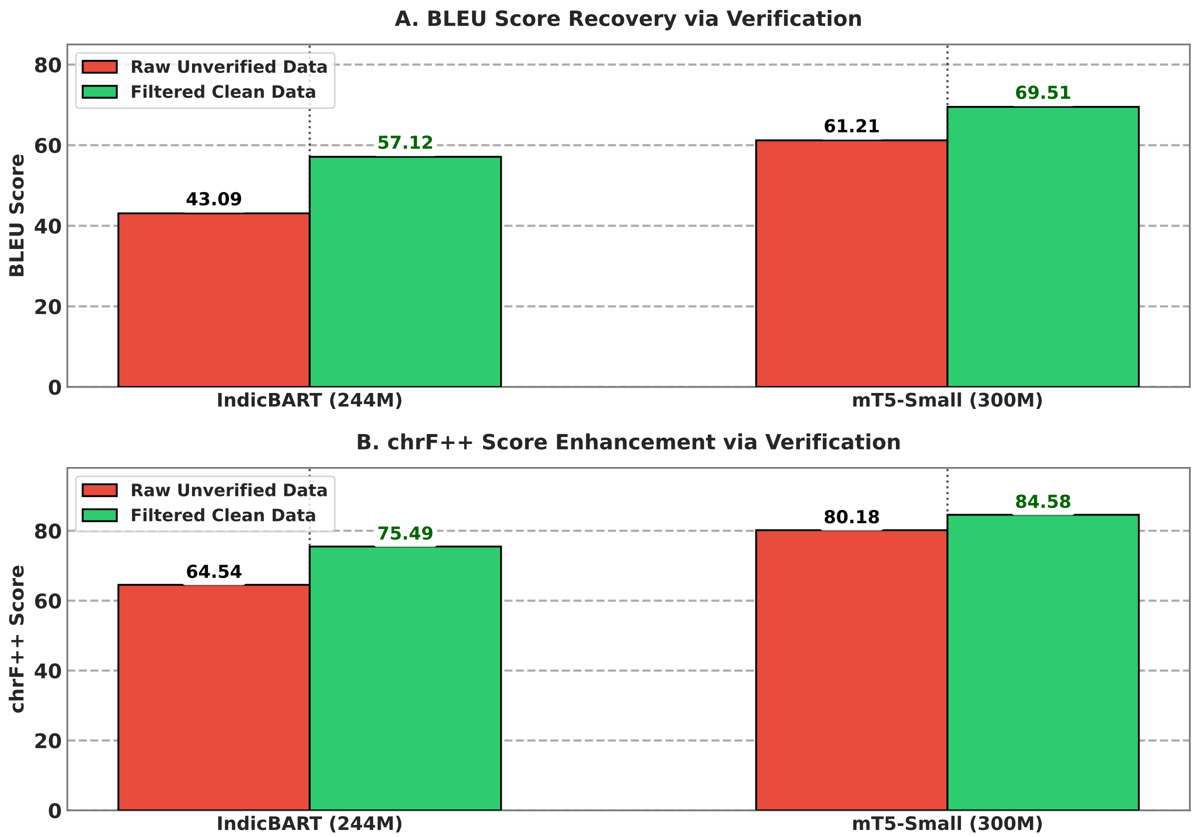

Figure 8: Multi-tier verification engine ablation: (A) BLEU score recovery and (B) chrF++ score enhancement comparing models trained on raw unverified synthetic outputs vs. filtered clean synthetic data.

5 Conclusion

This study presents a scalable, reproducible framework for multi-dialect Marathi text normalization, addressing the critical scarcity of parallel corpora for low-resource Indic varieties. By replacing expensive manual annotations with a synthetic parallel generation pipeline combining Gemma-4 translation with a multi-tier verification engine (rule-based script filtering and LLaMA-3.1-8B semantic auditing), we successfully expanded the clean parallel corpus from 16,163 to 32,335 verified pairs across Southern Konkan, Northern Konkan, and Varhadi dialects. Empirical evaluations demonstrate that mT5-small achieves state-of-the-art performance with 69.51 BLEU and 84.58 chrF++ on the expanded multi-dialect dataset (+12.39 BLEU over IndicBART). Crucially, multi-tier verification prevented synthetic quality collapse (+14.03 BLEU recovery over unverified data), while dataset expansion mitigated cross-dialect parameter competition and maintained near-perfect Standard Marathi normalization (97.39 BLEU, 1.37% WER) on held-out RESPIN benchmarks.

While these findings establish a strong blueprint for Indic sub-dialect normalization, minor constraints remain: LLM-driven verification pipelines introduce computational and latency overheads, synthetic expansion can occasionally yield slight precision dilution on already high-performing dialects like Varhadi, and intrinsic text metrics should be complemented by downstream task evaluations (e.g., ASR post-processing) in future work. Overall, our verified synthetic pipeline provides a scalable foundation for expanding digital linguistic inclusion across low-resource regional dialects.

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